SKEMA JAWAPAN DAN PEMARKAHAN PRA PSPM 3, FIZIK SEMESTER 1, SESI 2023/2024

<table>
<tr><th>NO.</th><th>ANSWER SCHEME</th><th>MARK (S)</th></tr>
<tr><td>1.(a)</td><td>

V = 15 cm^3 = 15×(10^{-2})^3 = 1.5×10^{-5} m^3
m = 27 g = 0.027 kg

$\rho = \frac{m}{V}$

$\rho = \frac{0.027}{1.5\times10^{-5}}$

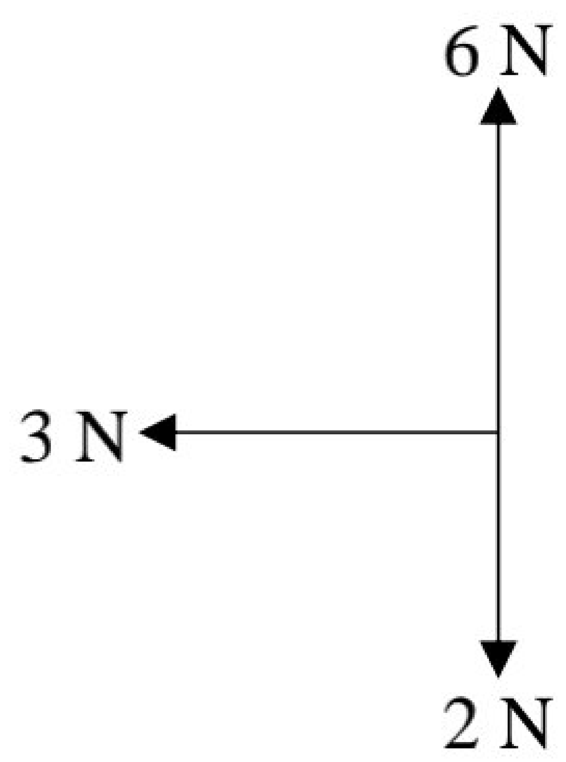
$\rho = 1800\,\text{kg m}^{-3}$

</td><td>

JU1

JU1

</td></tr>
<tr><td>1.(b)</td><td>



<table>
<tr><th></th><th>x-component (N)</th><th>y-component (N)</th></tr>
<tr><td>F$_1$</td><td>0</td><td>6</td></tr>
<tr><td>F$_2$</td><td>0</td><td>–2</td></tr>
<tr><td>F$_3$</td><td>–3</td><td>0</td></tr>
<tr><td>$\sum F$</td><td>$\sum F_x = -3$</td><td>$\sum F_y = 4$</td></tr>
<tr><td></td><td>G1</td><td>G1</td></tr>
</table>

$F = \sqrt{F_x^2 + F_y^2}$

$F = \sqrt{(-3)^2 + 4^2}$

$F = 5\,\text{N}$

$\theta = \tan^{-1}\left(\frac{\sum F_y}{\sum F_x}\right)$

$\theta = \tan^{-1}\left(\frac{4}{-3}\right)$

$\theta = -53.13^o$ (above the negative x-axis) /
$\theta = 53.13^o$ (positive answer also accepted)

</td><td>

G2

K1

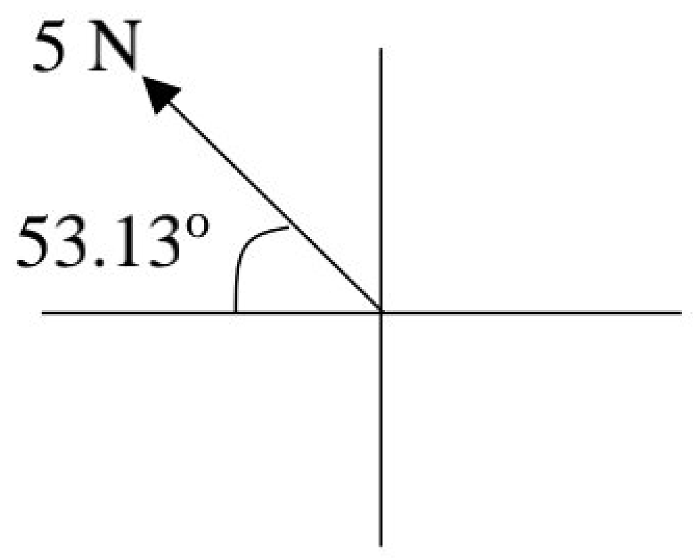
GJU1

K1

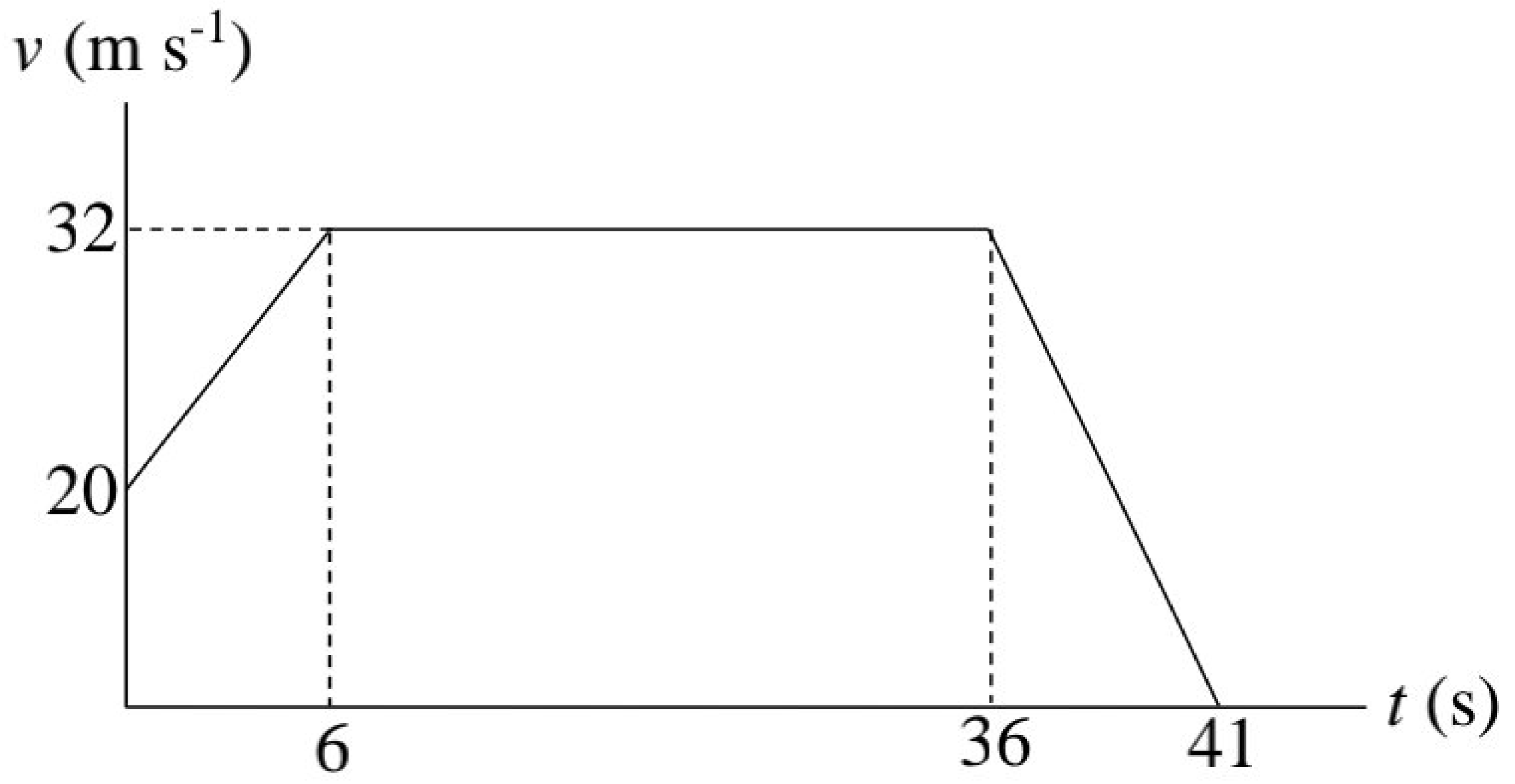
GJU1

</td></tr>
</table>

SKEMA JAWAPAN DAN PEMARKAHAN PRA PSPM 3, FIZIK SEMESTER 1, SESI 2023/2024

NO.	ANSWER SCHEME	MARK (S)
1.(b)	**OR** 	
	TOTAL	**8**

SKEMA JAWAPAN DAN PEMARKAHAN PRA PSPM 3, FIZIK SEMESTER 1, SESI 2023/2024

NO.	ANSWER SCHEME	MARK (S)
2.(a)(i)	$u = 20$ m s^{-1} $a = 2$ m s^{-2} $t = 6$ s $v = u + at$ $v = 20 + (2)(6)$ $v = 32$ m s^{-1}	 **G1** **JU1**
2.(a)(ii)	**<u>Graph v versus t</u>**  Label value of v&t and label the symbol and unit of v&t : 1 mark Shape : 1 mark	 **D2**
2.(a)(iii)	Total distance = area under v-t graph $\text{Total distance} = \frac{1}{2}\times(20+32)\times 6 + \frac{1}{2}\times(35+30)\times 32$ $\text{Total distance} = 1196\ \text{m}$	**K1** **G1** **JU1**
2.(b)(i)	$u = 12$ m s^{-1} $v = 0$ m s^{-1} $v^2 = u^2 - 2gs$ $0^2 = 12^2 - 2(9.81)s$ $s = 7.34\ \text{m}$	 **K1** **G1** **JU1**
2.(b)(ii)	$s = 0$ m $s = ut - \frac{1}{2}gt^2$ $0 = (12)t - \frac{1}{2}(9.81)t^2$ $t = 2.45\ \text{s}$	 **G1** **JU1**
	TOTAL	**12**

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NO.	ANSWER SCHEME	MARK (S)
3.(a)(i)	m = 0.57 kg If $u = +21$ m s^{-1}, $J = -20.5$ N s $J = m(v-u)$ $-20.5 = 0.57(v-21)$ $v = -14.96$ m s^{-1} **OR** If $u = -21$ m s^{-1}, $J = +20.5$ N s $J = m(v-u)$ $20.5 = 0.57(v-(-21))$ $v = 14.96$ m s^{-1}	 **G1** **JU1**
3.(a)(ii)	$J = Ft$ $20.5 = F(0.6)$ $F = 34.17$ N	 **G1** **JU1**
3.(b)	$m_1 = m$, $m_2 = m$ $u_1 = 5$ m s^{-1}, $u_2 = 0$ m s^{-1} $u_{1x} = 5$ m s^{-1}, $u_{2x} = 0$ m s^{-1} $u_{1y} = 0$ m s^{-1}, $u_{2y} = 0$ m s^{-1} $v_{1x} = v_1\cos32^\circ = 0.85v_1$, $v_{2x} = v_2\cos51^\circ = 0.63v_2$ $v_{1y} = v_1\sin32^\circ = 0.53v_1$, $v_{2y} = -v_2\sin51^\circ = -0.78v_2$ $\sum P_i = \sum P_f$ ***x*-component** $\sum P_{ix} = \sum P_{fx}$ $m_1u_{1x} + m_2u_{2x} = m_1v_{1x} + m_2v_{2x}$ $\cancel{m}(5) + \cancel{m}(0) = \cancel{m}(0.85v_1) + \cancel{m}(0.63v_2)$ $0.85v_1 + 0.63v_2 = 5$ ***y*-component** $\sum P_{iy} = \sum P_{fy}$ $m_1u_{1y} + m_2u_{2y} = m_1v_{1y} + m_2v_{2y}$ $\cancel{m}(0) + \cancel{m}(0) = \cancel{m}(0.53v_1) + \cancel{m}(-0.78v_2)$ $0.53v_1 - 0.78v_2 = 0$ $v_1 = 3.91$ m s^{-1} $v_2 = 2.66$ m s^{-1}	 **K1** **G1** **G1** **JU1** **JU1**
	TOTAL	**9**

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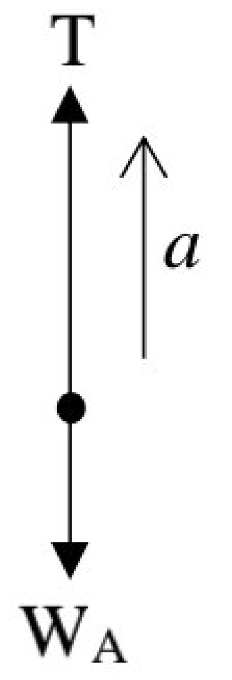
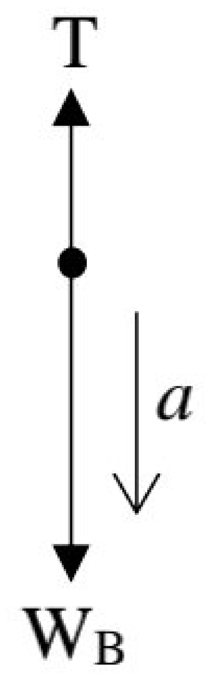
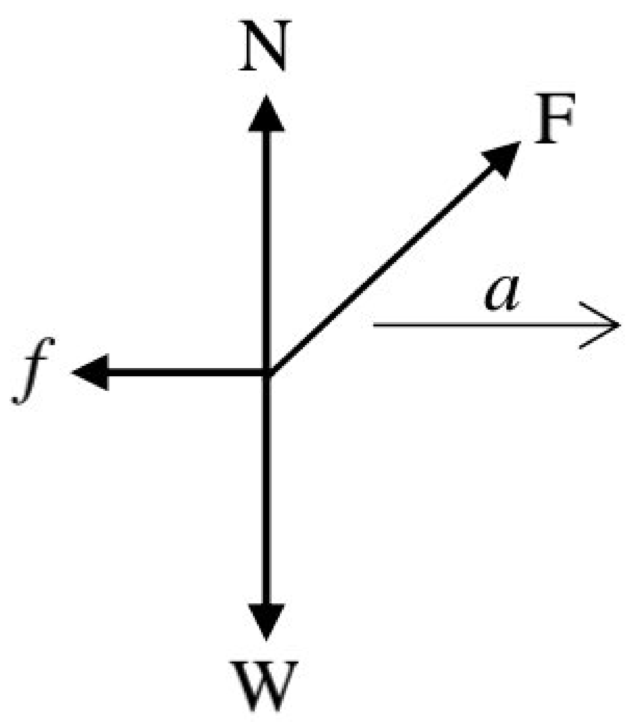
NO.	ANSWER SCHEME	MARK (S)
4.(a)	**FBD Object A**  **FBD Object B**  $m_A = 250$ g $= 0.25$ kg $m_B = 2m_A = 2\times 0.25 = 0.5$ kg $\sum F_y = m_A a$ $T - W_A = m_A a$ $T - (0.25)(9.81) = 0.25a$ $T - 2.45 = 0.25a$ $\sum F_y = m_B a$ $W_B - T = m_B a$ $(9.81)(0.5) - T = 0.5a$ $4.91 - T = 0.5a$ T = 3.27 N $a = 3.28$ m s^{-2}	**K1** **G1** **G1** **JU1** **JU1**
4.(b)(i)	 4 corrects : 2 marks 3 corrects : 1 mark (0 – 2) correct/s : 0 mark	**D2**

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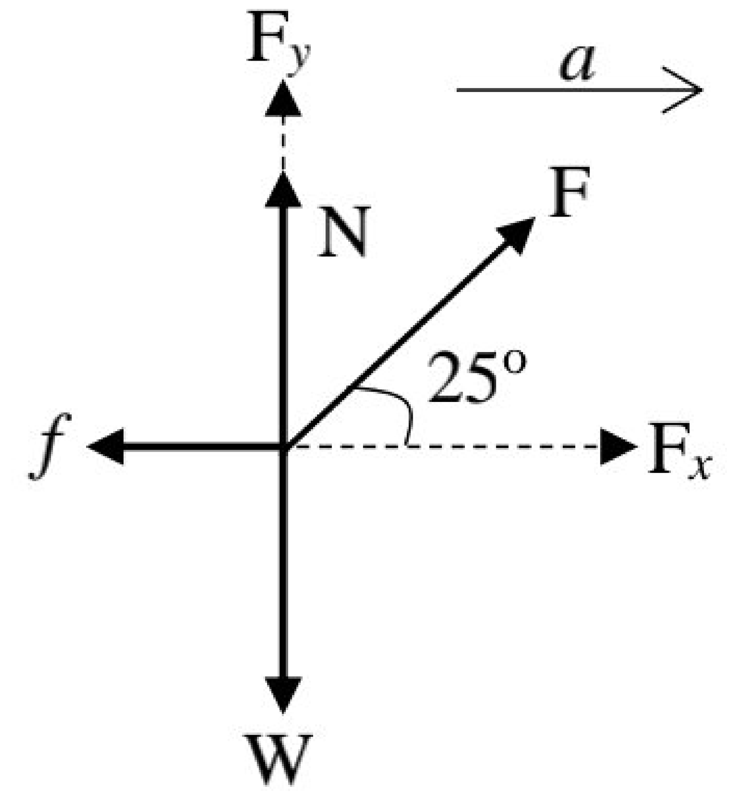
NO.	ANSWER SCHEME	MARK (S)
4.(b)(ii)	 	
	$F_x = F\cos 25^o = 0.91F$ $F_y = F\sin 25^o = 0.42F$	
	Constant speed $\rightarrow a = 0$ m s^{-2} $\rightarrow \sum F = 0$	K1
	$\sum F_y = 0$ $F_y + N - W = 0$ $0.42F + N = 680$	G1
	$f = \mu N$ $f = 0.5N$	
	$\sum F_x = 0$ $F_x - f = 0$ $0.91F - 0.5N = 0$	G1
	$F = 303.57$ N $N = 552.5$ N	JU1 JU1
	TOTAL	**12**

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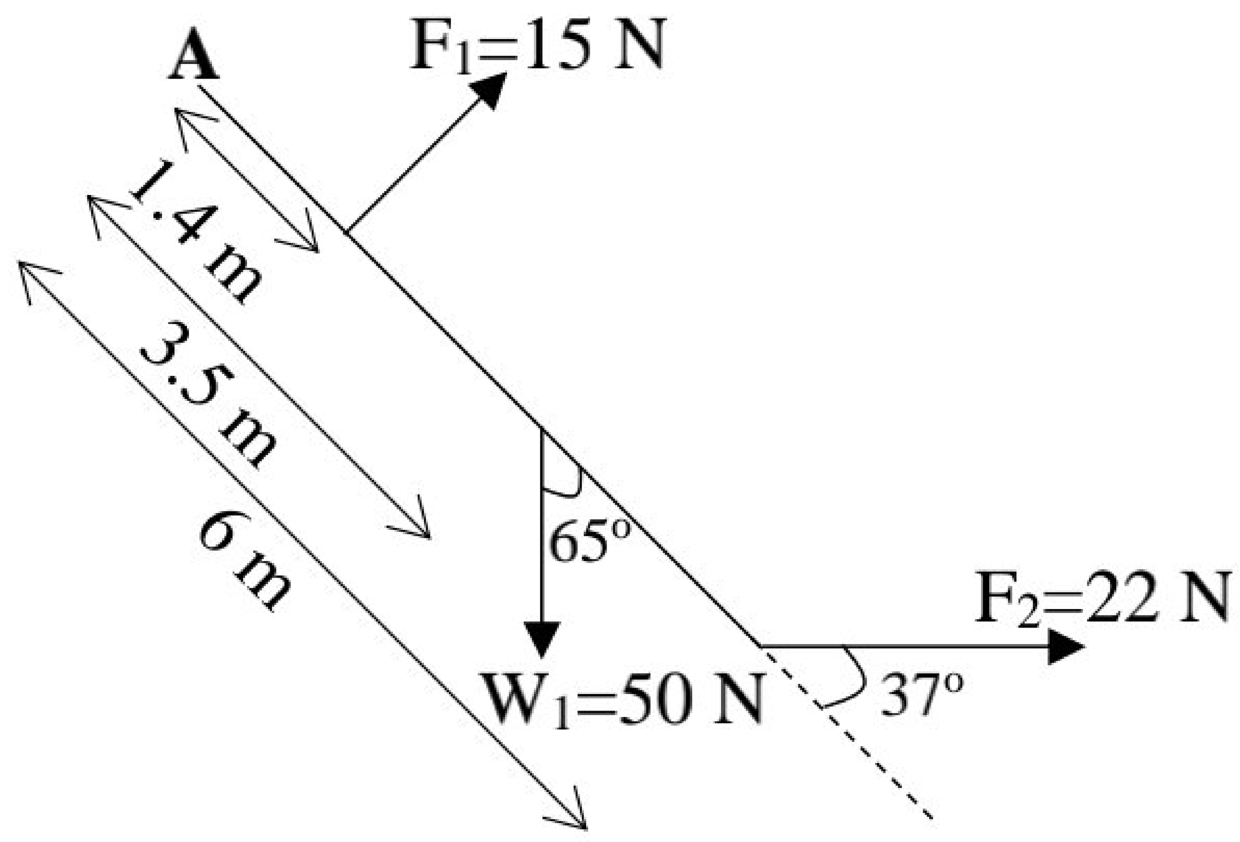
NO.	ANSWER SCHEME	MARK (S)
5.(a)(i)	m = 60 500 kg, $r=\frac{d}{2}=\frac{1\times10^5}{2}=5\times10^4\text{ m}$	
	$\omega=\frac{2\text{ rotation}}{3.6\times10^3}=\frac{2\times2\pi\text{ rad}}{3.6\times10^3\text{ s}}=3.49\times10^{-3}\text{ rad s}^{-1}$	**G1**
	$v=r\omega=(5\times10^4)(3.49\times10^{-3})$	**G1**
	$v=174.5\text{ m s}^{-1}$	**JU1**
5.(a)(ii)	$\omega=\frac{2\pi}{T}$	**K1**
	$3.49\times10^{-3}=\frac{2\pi}{T}$	
	$T=1800.34\text{ s}$	**GJU1**
5.(b)(i)	m = 50 kg, r = 2 m, $\omega=3\text{ rad s}^{-1}$	
	$a_c=r\omega^2$	
	$a_c=(2)(3)^2$	**G1**
	$a_c=18\text{ m s}^{-2}$	**JU1**
5.(b)(ii)	$F_c=ma_c$	
	$F_c=(50)(18)$	**G1**
	$F_c=900\text{ N}$	**JU1**
	TOTAL	**9**

SKEMA JAWAPAN DAN PEMARKAHAN PRA PSPM 3, FIZIK SEMESTER 1, SESI 2023/2024

NO.	ANSWER SCHEME	MARK (S)
6.(a)(i)	$\omega_o = \frac{5 \text{ rev}}{\text{min}} = \frac{5\times 2\pi \text{ rad}}{60 \text{ s}} = \frac{\pi}{6}$ rad s^{-1} / 0.52 rad s^{-1} $\omega = \frac{11 \text{ rev}}{\text{min}} = \frac{11\times 2\pi \text{ rad}}{60 \text{ s}} = \frac{11\pi}{30}$ rad s^{-1} / 1.15 rad s^{-1} $t = 100\,\text{s}$ $r = 0.08\,\text{m}$ $\omega = \omega_o + \alpha t$ $1.15 = 0.52 + \alpha(100)$ $\alpha = 6.3\times 10^{-3}$ rad s^{-2}	 G1 JU1
6.(b)(ii)	$a_t = r\alpha$ $a_t = (0.08)(6.3\times 10^{-3})$ $a_t = 5.04\times 10^{-4}$ m s^{-2} (**ecf** for α)	 G1 JU1
6.(b)(iii)	$\omega^2 = \omega_o^2 + 2\alpha\theta$ $1.15^2 = 0.52^2 + 2(6.3\times 10^{-3})\theta$ $\theta = 83.5\,\text{rad}$ $\frac{x \text{ rev}}{1 \text{ rev}} = \frac{83.5 \text{ rad}}{2\pi \text{ rad}}$ $x = 13.29$ rev/turns	 G1 K1 JU1
6.(b)(i)	 $\sum\tau = \tau_{acw} + (-\tau_{cw})$ $\sum\tau = (1.4)((15)\sin 90^o + (6)(22)\sin 37^o - (3.5)(50)\sin 65^o$ $= -58.16\,\text{N m}$ (clockwise)	 K1 G1 JU1

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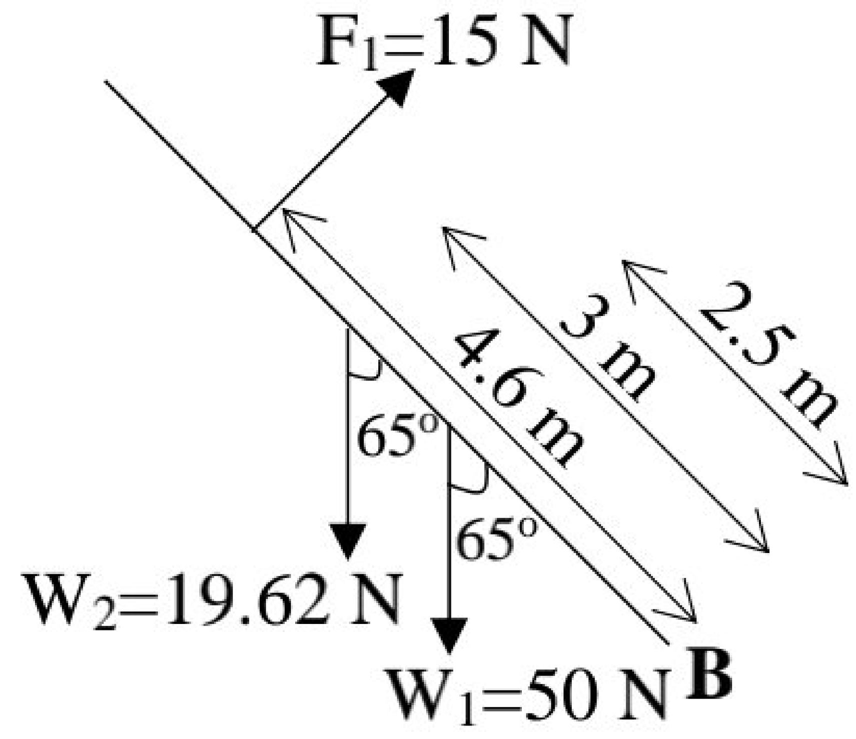
NO.	ANSWER SCHEME	MARK (S)
6.(b)(ii)		
	$W_2 = mg = (2)(9.81) = 19.62\,\text{N}$	
	$\sum \tau = \tau_{acw} + (-\tau_{cw})$	
	$\sum \tau = (3)(19.62)\sin 65^o + (2.5)(50)\sin 65^o - (4.6)(15)\sin 90^o$	G1
	$\sum \tau = 97.63\,\text{N m}$ (anticlockwise/counterclockwise)	JU1
	TOTAL	**12**

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NO.	ANSWER SCHEME	MARK (S)
7.(a)(i)	$A = 0.1\times10^{-4}$ cm^2 = 1×10^{-5} m^2, $L = 4\times10^{-3}$ m , T_1 = 373 K, T_2 = 223 K, k = 80 W m^{-1} K^{-1} $\frac{\Delta T}{L}=\frac{223-373}{4\times10^{-3}}$ $\frac{\Delta T}{L}=-37500$ K m^{-1}	 **G1** **JU1**
7.(a)(ii)	$\frac{Q}{t}=-kA\frac{\Delta T}{L}$ $\frac{Q}{24\times60\times60}=-(80)(1\times10^{-5})(-37500)$ $Q=2.59\times10^{6}$ J (**ecf** for $\frac{\Delta T}{L}$)	 **G1** **JU1**
7.(b)	$M_H = 2\times10^{-3}$ kg mol^{-1} $M_{O2} = 32\times10^{-3}$ kg mol^{-1} T_H = 47+273.15 K = 320.15 K $v_{rms,H}=\sqrt{\frac{3RT_H}{M_H}}$ $v_{rms,H}=\sqrt{\frac{3\times8.31\times320.15}{2\times10^{-3}}}=1997.67$ m s^{-1} $v_{rms,H}=v_{rms,O_2}=1997.67$ m s^{-1} $v_{rms,O_2}=\sqrt{\frac{3RT_{O_2}}{M_{O_2}}}$ $1997.67=\sqrt{\frac{3\times8.31\times T_{O_2}}{32\times10^{-3}}}$ $T_{O_2}=5122.42$ K	 **G1** **G1** **JU1**

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NO.	ANSWER SCHEME	MARK (S)
7.(c)(i)	Q = −125 kJ W = −104 kJ $Q = \Delta U + W$ $-125k = \Delta U + (-104k)$ $\Delta U = -21\,\text{kJ}$	 **G1** **JU1**
7.(c)(ii)	Q = 260 kJ ΔU = 157 kJ $Q = \Delta U + W$ $260k = 157k + W$ $W = 103\,\text{kJ}$	 **G1** **JU1**

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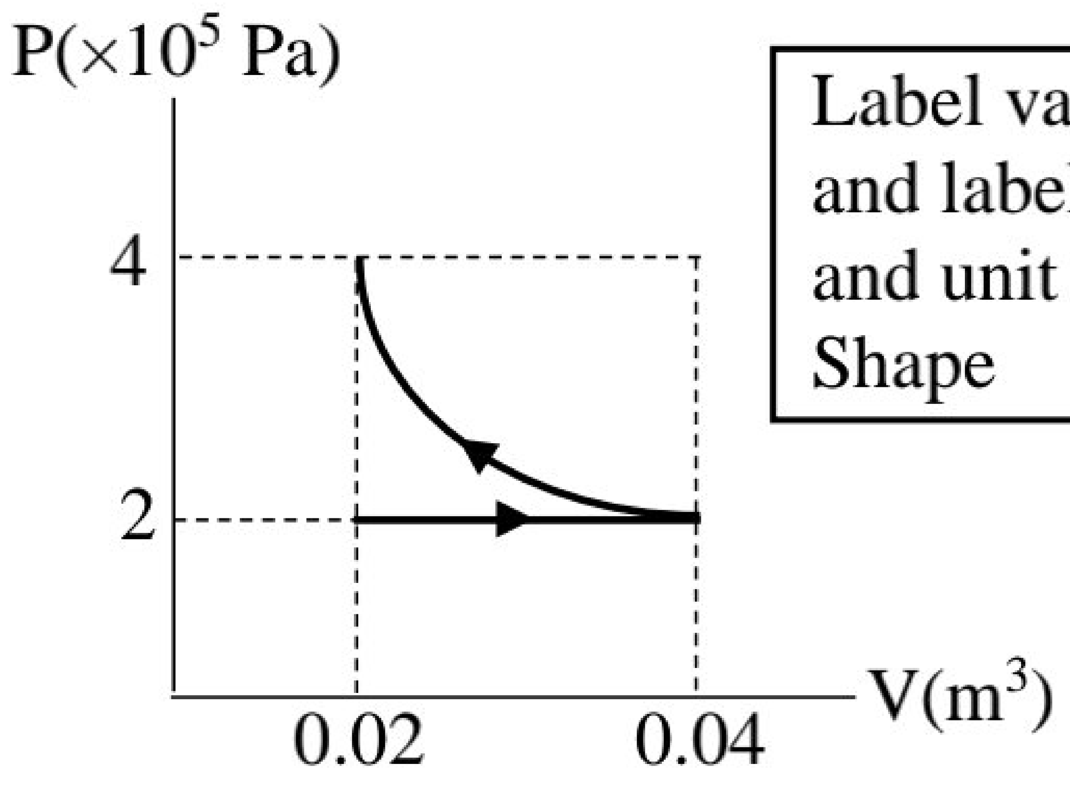
NO.	ANSWER SCHEME	MARK (S)
7.(d)(i)	n = 2 moles $\frac{V_1}{T_1}=\frac{V_2}{T_2} \rightarrow \frac{0.02}{(27+273.15)}=\frac{0.04}{T_2} \rightarrow T_2 = 600.3\,\text{K}$ [T_1 = 300.15 K, V_1 = 0.02 m^3, P_1 = 2×10^5 Pa] $\xrightarrow[\text{W = 4000 J, } \Delta\text{U = 7483 J}]{\text{Constant pressure (Isobaric process)}}$ [T_2 = 600.3 K, V_2 = 0.04 m^3, P_2 = 2×10^5 Pa] $\xrightarrow[\text{Q = -6916 J}]{\text{Constant temperature (Isothermal compression process)}}$ [T_3 = 600.3 K, V_3 = 0.02 m^3, P_3 = 4×10^5 Pa]  Label value of P&V : 1 mark and label the symbol and unit of P&V Shape : 1 mark	D2
7.(d)(ii)	$W = +4000\,\text{J}$ $\Delta U = +7483\,\text{J}$ $Q = \Delta U + W$ $Q = 7483 + 4000$ $Q = 11483\,\text{J}$ Heat is absorbed.	 G1 JU1 J1
7.(d)(iii)	Isothermal process → ΔU = 0 J Q = −6916 J $Q = \Delta U + W$ $-6916 = 0 + W$ $W = -6916\,\text{J}$ Work is done on the system.	 G1 JU1 J1
	TOTAL	**18**